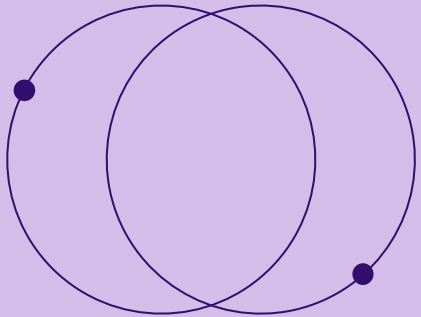




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Genex Services

Genex Services is a leading provider of integrated managed care solutions for the workers' compensation industry dedicated to enhancing the results of wounded workers and minimizing expenses for insurance companies and employers, Genex provides a full range of services that are customized to satisfy the various demands of its customers.

Services Provided:

Case Management: To provide prompt and efficient medical care and return-to-work options, Genex's team of skilled case managers collaborates closely with injured workers, healthcare providers, and stakeholders.

Utilization Review: Genex thoroughly examines medical care requests to make sure they are suitable and cost-effective, using cutting-edge technology and evidence-based criteria.

Medical Provider Network: Genex upholds a national network of accredited healthcare providers, providing injured workers with access to high-quality care while encouraging cost-cutting and effective claims handling.

Telehealth Services: Genex offers easy access to medical knowledge through virtual consultations and remote monitoring, enhancing access to care and cutting down on needless travel and waiting times.

Data Analytics: Genex uses analytics and big data to provide insights and predictive modeling for trend detection, risk reduction, and optimizing outcomes for clients.

Branch Setup:

Genex Services LLC does business through a network of strategically placed regional branches spread throughout important geographical areas. A committed team of specialists, comprising case managers, medical professionals, and administrative personnel, work in each office to offer customers and injured workers customized assistance and services.

Staffing: Genex has a multidisciplinary team of specialists with experience in business administration, technology, insurance, and healthcare. Each branch is usually staffed to a sufficient degree to meet customer expectations and guarantee the provision of high-quality services; however, the precise number of employees differs depending on state's requirements and branch size. Overall, Genex Services has 2,075 employees.

Genex Security and Framework

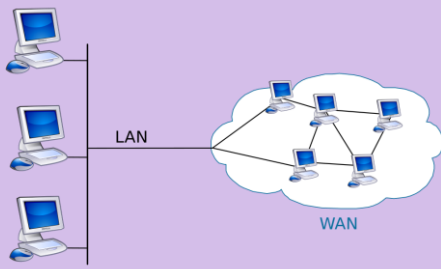
1. At layer one, the physical layer there are sniffing attacks. To prevent these attacks, our company will take measures by using endpoint security, implementing network access controls, and regular security audits.
2. At layer two, the data link layer there are spoofing attacks. To prevent spoofing attacks, our company will implement multi-factor authentication, and digital signatures and implement DNSSEC to protect against DNS spoofing attacks.
3. At layer three, the network layer is popular for man-in-the-middle attacks. To protect against this we will use secure communication protocols such as HTTPS for web-based applications and secure email protocols. We will also use VPNs to create encrypted tunnels, two-factor authentication, and implement strong Wi-Fi security protocols, and use complex passwords.
4. Layer 4 is the transport layer and reconnaissance/DoS attacks happen in this layer. To prevent these attacks our company will conduct regular security audits along with deploying IDS to monitor the network traffic and detecting suspicious behavior, and Load Balancing will be used to distribute traffic across multiple servers.
5. Layer 5, is the session layer and hijacking is a concern at this layer. To address this concern we will implement session timeout mechanisms along with using secure cookies and mutual authentication.
6. Layer 6, is the presentation layer and phishing is a concern at this layer. To address this concern we will use content filtering, user training, and awareness and implement URL rewriting to modify and inspect URLs in emails, making them safer for users to click.
7. Layer 7, is the application layer, and exploiting is a major concern. To keep our company safe from this we will promote secure coding practices, deploy web application firewalls, use secure session management practices, and implement monitoring services.

Why the NIST Framework?

- **Widely Recognized:** The NIST Framework is globally accepted and endorsed by leading organizations and regulatory bodies.
- **Tailored Approach:** NIST offers a flexible and scalable framework that can be customized for the challenges and regulatory requirements of the healthcare industry.
- **Complete Coverage:** The framework includes five functions - Identify, Protect, Detect, Respond, and Recover - providing a complete approach to managing cybersecurity risks.
- **Risk-Based Methodology:** NIST promotes a risk-based approach to cybersecurity, allowing Genex to prioritize focus based on the potential impact on business operations and sensitive data.
- **Integration with Standards:** NIST aligns with industry standards such as HIPAA, GDPR, and ISO 27001, ensuring compliance and compatibility with existing regulatory initiatives.
- **Continuous Improvement:** The NIST Framework promotes continuous improvement through regular assessments, monitoring, and adaptation to evolving threats and technologies.

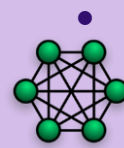


Genex Network



- **Wide Area Network (WAN)**
- **Connectivity Across Locations:** Enables communication and data exchange between employees across 38 different locations.
- **Scalability:** Accommodates additional sites and users as the company grows, adapting to size and location changes.
- **Centralized Management:** Facilitates centralized management of network resources and policies, ensuring efficiency and reducing administrative overhead.
- **Resource Sharing and Collaboration:** Allows for sharing resources such as files, applications, and databases, allowing collaboration and teamwork across locations.
- **Redundancy and Reliability:** Provides built-in redundancy and failover mechanisms to ensure network reliability and minimize downtime.

WAN is the most suitable network type for Genex network infrastructure. It provides the necessary connectivity, scalability, centralized management, resource sharing, and reliability to support its operations effectively across 38 locations and over 2000 employees.



- **Logical Topology: Mesh**

- Provides redundant paths between locations for high reliability and fault tolerance.
- Ensures continuous connectivity in case of link failures or network outages.



- **Physical Topology: Star**

- Centralizes connections at a central hub or switch for simplified management and troubleshooting.
- Easy expansion and scalability by connecting new locations to the central hub.



- **Logical Topology: Virtual Private Network (VPN)**

- Establishes secure and encrypted connections over public networks for remote locations.
- Enables secure access to company resources and data from any location while maintaining confidentiality.



- **Physical Topology: Ethernet**

- Utilizes twisted-pair or fiber-optic cables for high-speed and reliable connectivity within each location.
- Supports efficient internal communication and data exchange, facilitating day-to-day operations.



- **Logical Topology: Point-to-Point Protocol (PPP)**

- Direct connections between nodes over serial links for remote locations.
- Provides dedicated and secure communication channels, ensuring reliable connectivity.

The combination of mesh, star, VPN, Ethernet, and PPP topologies ensures reliable, secure, and efficient network connectivity and communication across 38 locations and 2075 employees. Each topology plays a specific role in providing redundancy, scalability, security, and performance to support the organization's network infrastructure and operations.